

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Science in Information Technology (M.Sc.(IT))
Semester-III University Examination November 2018
MS 317 Web Development using Advanced Java

Date: 24.11.18, Saturday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION - I**Q1 Do as directed. [11]**

1. _____ package in java is used to support JDBC API.
2. Which of the following manages a list of database drivers in JDBC?
a. DriverManager b. Connection
3. In _____ type of ResultSet, the cursor cannot scroll backwards.
a. ResultSet.TYPE_SCROLL_INSENSITIVE b. ResultSet.TYPE_FORWARD_ONLY
4. Servlet embeds HTML in java. State True/False.
5. When web container gets the request and if it's for servlet then container creates two Objects HttpServletRequest and HttpServletResponse. State True/False.
6. Files stored in _____ folder under your directory structure are not supposed to be accessible from a browser.
a. WEB-INF b. WEB PAGES
7. _____ form of execute method is suitable for a select query.
8. What is the full form of MIME ?
9. Which Writer class is used to write the response from Servlet ?
a. FileWriter b. PrintWriter
10. Servlets handle multiple simultaneous requests by using threads. State True/False.
11. The _____ API lets you define HTTP-specific classes in java.

Q2 Answer the following questions.**[A]** Explain the Servlet life cycle with suitable diagram. Explain each method of life cycle in brief. **[06]****[B]** Explain any six methods of ResultSet Interface with example. **[06]****OR****Q2 Answer the following questions.****[A]** List down the steps to configure JDBC for a simple java application. Explain them briefly. **[06]****[B]** Write a code in Servlet to read and display four request parameters namely, firstname, lastname, email and mobilenumber, submitted by a form. **[06]****Q3 Answer the following questions. (Any Three) [12]****[A]** Explain the methods to register and connect the database using JDBC.**[B]** Explain Statement and PreparedStatement with example.**[C]** What is ResultSetMetaData? Explain any three methods provided by the ResultSetMetaData interface.**[D]** What is JDBC Transaction? Explain the methods for performing commit and rollback.**[E]** Explain the page directive with any four attributes in detail.

SECTION – II

Q4 Do as directed. [11]

1. `_jspService()` method of `HttpJspPage` class should not be overridden. State True/False.
2. The implicit object “application” is instance of which class?
a. `javax.servlet.Application` b. `javax.servlet.ServletContext`
3. The default content type of JSP is _____.
a. `text/html` b. `text/plain`
4. _____ syntax is used to write a jsp declaration.
5. Which tag is used to write and execute java source code in JSP?
a. expression tag b. scriptlet tag
6. _____ is the default scope in JSP when the scope attribute is omitted.
7. JSP's eventually are compiled into Java Servlets, you can do as much with JSP's as you can do with Java Servlets. State True/False.
8. Which of the following attributes are used in `jsp:include` tag?
a. `id,type` b. `page,flush`
9. JSTL stands for _____.
10. Which attribute is used in `c:forTokens` to separate the list in tokens?
a. `items` b. `delims`
11. _____ tag can include the content, even if the content resides on a different Web server.
a. `<c:url>` b. `<c:import>`

Q5 Answer the following questions.

- [A] What is the role of HTTP in client server communication? Explain the features of HTTP and list the category of HTTP status codes. [06]
- [B] Write the steps of Implementing MVC with `RequestDispatcher`. [06]

OR

Q5 Answer the following questions.

- [A] Create an `ArrayList` of employee names in a Servlet, share the list for the current request and access the list in a JSP view using the Expression Language. [06]
- [B] Explain the syntax of `jsp:useBean`, `jsp:getProperty` and `jsp:setProperty`. [06]

Q6 Answer the following questions. (Any Three)

- [A] Write a code to map the Servlet name, Class and Url in `web.xml` file. [12]
- [B] Differentiate between `forward` and `sendRedirect` method.
- [C] Explain the syntax and usage of JSP Scriptlet and Expression.
- [D] Write and explain the syntax to access the implicit objects and bean properties using Expression Language.
- [E] Explain `c:set` and `c:remove` tags of JSTL.